

Lecture-4

Mathematics 1 (15B11MA111)

Module: Partial Differentiation

CO1 [C105.1]

Topics to be covered

➤ Partial Derivatives

- ❖ Definition

- ❖ Mathematical concept

- ❖ Examples

- ❖ Assignment

➤ Reference for the lecture

R.K Jain and S.R.K. Iyenger, “Advanced Engineering Mathematics” fifth edition, Narosa publishing house, 2016.

Partial Derivatives

➤ Definition:

If ' f ' is a function of two or more variables then the derivative of function ' f ' with respect to one of its independent variables keeping all other independent variables as constant is called the partial derivative of the function with respect to that variable.

Mathematical Concept

Consider the function of two variables $z = f(x, y)$ defined in some domain D of the x - y plane.

Let y be held constant, say $y = y_0$ then the **first order partial derivative of z or $f(x, y)$ with respect to x** at the point (x_0, y_0) is given by

$$\begin{aligned}\left. \frac{\partial z}{\partial x} \right|_{(x_0, y_0)} &= \left. \frac{\partial f}{\partial x} \right|_{(x_0, y_0)} = \lim_{\Delta x \rightarrow 0} \frac{\Delta_x z}{\Delta x} \\ &= \lim_{\Delta x \rightarrow 0} \frac{f(x_0 + \Delta x, y_0) - f(x_0, y_0)}{\Delta x}\end{aligned}$$

provided the limit exists and is denoted by $z_x(x_0, y_0)$ or $f_x(x_0, y_0)$.

Similarly, if x be held constant, say $x = x_0$ then the **first order partial derivative of z or $f(x, y)$ with respect to y** at the point (x_0, y_0) is given by

$$\begin{aligned}\left. \frac{\partial z}{\partial y} \right|_{(x_0, y_0)} &= \left. \frac{\partial f}{\partial y} \right|_{(x_0, y_0)} = \lim_{\Delta y \rightarrow 0} \frac{\Delta_y z}{\Delta y} \\ &= \lim_{\Delta y \rightarrow 0} \frac{f(x_0, y_0 + \Delta y) - f(x_0, y_0)}{\Delta y}\end{aligned}$$

provided the limit exists and is denoted by $z_y(x_0, y_0)$ or $f_y(x_0, y_0)$.

Generalization of Partial Derivatives

Let $z = f(x_1, x_2, \dots, x_n)$ be a function of n variables defined in some domain D in $\mathcal{R}^n$. Let $P_0(x_1, x_2, \dots, x_n)$ be a point in D . If the limit

$$\lim_{\Delta x_i \rightarrow 0} \frac{\Delta_{x_i} z}{\Delta x_i} =$$
$$\lim_{\Delta x_i \rightarrow 0} \frac{f(x_1, x_2, \dots, x_i + \Delta x_i, \dots, x_n) - f(x_1, x_2, \dots, x_i, \dots, x_n)}{\Delta x_i}$$

exists, then it is called the partial derivative of f at the point P_0 and is denoted by $\left(\frac{\partial f}{\partial x_i}\right)_{P_0}$

Examples

Example 1: Find the first order partial derivatives of the function

$$f(x, y) = x^2 + y^2 + x$$

Solution: $f(x, y) = x^2 + y^2 + x$

$$\begin{aligned}\frac{\partial f}{\partial x} &= \lim_{\Delta x \rightarrow 0} \frac{f(x+\Delta x, y) - f(x, y)}{\Delta x} \\ &= \lim_{\Delta x \rightarrow 0} \frac{[(x+\Delta x)^2 + y^2 + (x+\Delta x)] - [x^2 + y^2 + x]}{\Delta x}\end{aligned}$$

$$= \lim_{\Delta x \rightarrow 0} \frac{(2x+1)\Delta x + (\Delta x)^2}{\Delta x}$$

$$= \lim_{\Delta x \rightarrow 0} [2x + 1 + \Delta x] = 2x + 1$$

Similarly, $\frac{\partial f}{\partial y} = \lim_{\Delta y \rightarrow 0} \frac{f(x, y + \Delta y) - f(x, y)}{\Delta y}$

$$= \lim_{\Delta y \rightarrow 0} \frac{[y^2 + (y + \Delta y)^2 + x] - [x^2 + y^2 + x]}{\Delta y}$$

$$= \lim_{\Delta y \rightarrow 0} \frac{2y \Delta y + (\Delta y)^2}{\Delta y}$$

$$= \lim_{\Delta y \rightarrow 0} [2y + \Delta y] = 2y$$

Example 2: Find the first order partial derivatives and their values at point (2,3) of the function

$$f(x, y) = ye^{-x}$$

Solution: $f(x, y) = ye^{-x}$

$$\frac{\partial f}{\partial x} = \lim_{\Delta x \rightarrow 0} \frac{f(x+\Delta x, y) - f(x, y)}{\Delta x}$$

$$\frac{\partial f}{\partial x} = \lim_{\Delta x \rightarrow 0} \frac{ye^{-(x+\Delta x)} - ye^{-x}}{\Delta x}$$

$$\frac{\partial f}{\partial x} = \lim_{\Delta x \rightarrow 0} \frac{-ye^{-x}(1 - e^{-\Delta x})}{\Delta x}$$

$$\frac{\partial f}{\partial x} = (-ye^{-x}) \lim_{\Delta x \rightarrow 0} \frac{(1-e^{-\Delta x})}{\Delta x}$$

$$\frac{\partial f}{\partial x} = -ye^{-x}$$

Similarly

$$\frac{\partial f}{\partial y} = \lim_{\Delta y \rightarrow 0} \frac{f(x, y+\Delta y) - f(x, y)}{\Delta y}$$

$$\frac{\partial f}{\partial y} = \lim_{\Delta y \rightarrow 0} \frac{(y+\Delta y)e^{-x} - ye^{-x}}{\Delta y}$$

$$\frac{\partial f}{\partial y} = \lim_{\Delta y \rightarrow 0} \frac{\Delta ye^{-x}}{\Delta y}$$

$$\frac{\partial f}{\partial y} = e^{-x}$$

Now we have to find values of partial derivatives at the point (2,3)

$$\left. \frac{\partial f}{\partial x} \right|_{(2,3)} = -3e^{-2} = -0.41045$$

$$\left. \frac{\partial f}{\partial y} \right|_{(2,3)} = e^{-2} = 0.13681$$

Example 3: Show that the function

$$f(x, y) = \begin{cases} \frac{xy}{x^2 + 2y^2}, & (x, y) \neq (0, 0) \\ 0, & (x, y) = (0, 0) \end{cases}$$

is not continuous at $(0,0)$ but its partial derivatives f_x and f_y exist at $(0,0)$.

Solution: Choose the path $y=mx$. Since the limit

$$\lim_{(x,y) \rightarrow (0,0)} f(x, y) = \lim_{x \rightarrow 0} \frac{mx^2}{(1+2m^2)x^2} = \frac{m}{(1+2m^2)}$$

depends on m , the function is not continuous at $(0, 0)$. We now have

$$f_x(0,0) = \left. \frac{\partial f}{\partial x} \right|_{(0,0)} = \lim_{\Delta x \rightarrow 0} \frac{f(\Delta x, 0) - f(0,0)}{\Delta x}$$

$$f_x(0,0) = \lim_{\Delta x \rightarrow 0} \frac{0-0}{\Delta x} = 0$$

$$f_y(0,0) = \left. \frac{\partial f}{\partial y} \right|_{(0,0)} = \lim_{\Delta y \rightarrow 0} \frac{f(0, \Delta y) - f(0,0)}{\Delta y}$$

$$f_y(0,0) = \lim_{\Delta y \rightarrow 0} \frac{0-0}{\Delta y} = 0$$

Therefore, the partial derivatives f_x and f_y exist at $(0,0)$.

Try Yourself

Q.1: For the function $f(x, y) = x^2 + 3xy + y - 1$, find first order partial derivatives.

Q.2: Show that the function

$$f(x, y) = \begin{cases} \frac{xy}{\sqrt{x^2 + y^2}}, & (x, y) \neq (0, 0) \\ 0, & (x, y) = (0, 0) \end{cases}$$

has partial derivatives $f_x(0, 0)$ and $f_y(0, 0)$, but the partial derivatives are not continuous at $(0, 0)$.

THANK YOU

